

Optipeople DemoCNC D-2800M — Installation Manual

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For factory technicians and qualified electrical contractors. Do not attempt installation without a copy of the latest commissioning checklist (D2800M-COMM-CHK-001) and at least one operator from the customer site present for the acceptance test.

1. Pre-delivery site survey

Complete **before** the machine ships. The site survey form (D2800M-SURV-001) must be returned to Optipeople at least 14 days before the agreed delivery date.

1.1 Building access

- Door / corridor / lift width must be $\geq 2,200$ mm.
- Floor along the route must be flat and free of ramps $> 5\%$.
- Crated machine: $6,100 \times 2,650 \times 2,150$ mm, **2,520 kg gross**. Plan a forklift with ≥ 3 t capacity and 1,800 mm forks.

1.2 Final position

- Floor flatness: ≤ 3 mm over the 5,820 mm machine length, measured with a 2 m straight-edge.
- Floor strength: $\geq 1,500$ kg/m² distributed load.
- Floor finish: sealed or epoxy-coated concrete recommended.
- Service clearances per drawing §10 of D2800M-DWG-001.

1.3 Utilities

Utility	Requirement
Electrical	380 V / 3-phase / 50 Hz, 80 A breaker, TN-S preferred
Compressed air	6 bar, 380 NL/min, ISO 8573-1 class 4-4-3, G 1/2" fitting
Vacuum	600 m ³ /h at -700 mbar (pump not included unless OP-OPT-VAC)
Dust extraction	5,800 m ³ /h at 2,500 Pa, 3× Ø 200 mm + 1× Ø 150 mm
Network	1× RJ-45 drop, gigabit, isolated VLAN preferred

2. Unpacking

1. Position the crate within 2 m of the final machine location.
2. Inspect the crate for shock-witness labels. If any indicator has triggered, photograph it and contact Optipeople before unpacking.
3. Remove the lid first, then the side panels. **Do not cut straps with the machine still bolted to the crate floor — the head assembly can shift when load is released.**
4. Verify the packing list against contents. Standard supply:

- Machine, base, integrated cabinet (1)
- Tool kit (1) — see §11
- Spare-parts kit OP-SPK-2800 (1) — see §11
- User documentation USB stick (1)
- Acceptance & training schedule (1)
- Lift sling, single-use (2)

3. Rigging & placement

1. Attach slings to the four red M20 lifting eyes on the upper frame corners. Sling angle $\geq 60^\circ$ from horizontal.
2. Lift slowly. Observe for binding on the safety enclosure — do not lift if any part of the enclosure is in contact with the frame.
3. Lower onto the four jacking pads. Final position must align with the pre-marked anchor pattern (drawing §8 of D2800M-DWG-001).
4. Level the machine by adjusting the eight footpads with a precision spirit level (≤ 0.05 mm/m) on the cross beam guideway. Check at eight positions along X. Tighten lock nuts to 80 Nm.
5. Mark and drill M16 chemical anchors. Cure per anchor manufacturer spec (typically 6–24 h at 20 °C) **before** energizing the machine.

4. Pneumatic connection

1. Confirm the shop air supply is dry and oil-free at the machine inlet.
2. Connect 6 bar supply to the G 1/2" inlet on the rear-left of the cabinet using the supplied flexible hose.
3. Open the manual ball valve V1 slowly. Verify the input pressure gauge reads 6.0 ± 0.2 bar.
4. With air on, listen for leaks at the FRL1 unit, manifolds M1–M3, and each presser cylinder. Soapy water test is acceptable.
5. Drain the FRL1 condensate bowl. The unit is fitted with an automatic float drain — confirm it cycles by briefly venting the supply.

5. Vacuum & dust connection

1. Connect the vacuum supply line ($\varnothing$ 125 mm, reinforced rubber) to the inlet flange under work zone A. Use the supplied clamp.
2. Connect the three $\varnothing$ 200 mm dust hoses to the spindle hoods, and the single $\varnothing$ 150 mm hose to the horizontal drilling head. Use band clamps, not zip-ties.
3. With the dust extractor running, measure flow at each port using an anemometer. Minimum face velocity at the hood opening: 25 m/s. If below, check for blockage or undersized main duct.

6. Electrical connection

⚠ Electrical work must be performed by a qualified electrician in accordance with EN 60204-1 and local regulations. Lock-out / tag-out the supply before opening the cabinet.

1. Verify supply is **dead** at the cabinet entry. Test with a verified voltmeter.
2. Route 4-core + PE cable through the cabinet's underside cable gland. Recommended cross-section: $4 \times 25 \text{ mm}^2 \text{ Cu}$ for $\leq 30 \text{ m}$ run length.

3. Connect L1, L2, L3 to terminals X1:1, X1:2, X1:3. Connect N to X1:4 only if a neutral is provided (TN-S). Connect PE to the green/yellow bus bar.
4. Verify phase rotation using the in-cabinet phase monitor K1. The green "PHASE OK" lamp must illuminate. If "PHASE ERR" lights, swap any two phases.
5. Energize. Verify:
 - 24 V DC bus reads 24.0 ± 0.5 V at TB-24V-1.
 - IPC boots and shows the OptiPanel splash.
 - Cabinet fan starts when internal temperature exceeds 35 °C.

7. Software commissioning

1. Sign in to the HMI as `installer` (initial password on the welcome sheet — change immediately).
2. Open **Settings** → **Machine** → **Identity** and confirm the serial number matches the nameplate.
3. Run **Settings** → **Diagnostics** → **I/O Live** and verify every door switch, E-stop, and pressure sensor.
4. Run **Settings** → **Diagnostics** → **Axis Home**. Each axis should home without alarm. Observe the home reference, then jog ± 50 mm and back to confirm position holds within ± 0.05 mm.
5. Run **Settings** → **Maintenance** → **Spindle Run-In**. The HMI cycles each spindle through 30 / 60 / 80 / 100 % speed for 10 minutes each. Do not skip — bearings need this to settle.
6. Calibrate tool length (vertical setter) and tool diameter (laser setter) for at least one tool in S1.

8. Mechanical commissioning

1. Belt tension (S3 horizontal milling head): plucked frequency should be 110 ± 5 Hz. Adjust the eccentric mount if outside.
2. Backlash test, X axis: command a 100 mm move in each direction and compare to a dial indicator. Result must be ≤ 0.03 mm. If higher, adjust the pinion preload.
3. Squareness (X to Y): mount a square on the table, sweep a 600 mm line along X with a dial indicator on the head. Sweep tolerance ≤ 0.05 mm.
4. Vacuum zone test: command **All Zones On**. Place a $\varnothing 200$ mm rubber pad in the middle of each zone and verify it cannot be pushed off by hand.

9. Acceptance test

The standard acceptance program is the file `acceptance_v3.opt` on the documentation USB. It runs a 30-minute mixed-operation job on a $600 \times 400 \times 18$ mm MDF blank. Pass criteria:

- All cuts within ± 0.10 mm of nominal.
- No spindle PT100 reading > 75 °C.
- No alarm or warning logged.
- Dust collection visually adequate at the cut zone.

The customer's appointed inspector signs the acceptance report at the HMI's **Service** → **Acceptance** screen. A PDF copy is written to the documentation USB and emailed to the address configured in **Settings** → **Account**.

10. Operator training

A factory-certified trainer delivers two days of on-site training. Day one covers safety, daily start-up/shutdown, panel loading, and running existing jobs. Day two covers tool setting, job creation in OptiPanel, import from MPR/DXF, and first-line troubleshooting.

The trainer signs off each operator on a printed competency card. Operators not signed off may not run the machine unsupervised.

11. Supplied kits

11.1 Tool kit OP-TK-2800

- Hex keys 2–17 mm (1 set)
- Torx keys T8–T40 (1 set)
- Open-end spanners 8–24 mm (1 set)
- ER32 tightening wrench set (2)
- ISO30 tool holder loading lever (1)
- Manual grease gun, 14 oz cartridge (1)
- Anemometer (1)
- Feeler gauge set 0.05–1.00 mm (1)

11.2 Spare parts kit OP-SPK-2800

- Pneumatic seal kit OP-SK-CYL (1)
- Vacuum gasket strip 6×2 mm × 5 m (1)
- Drive belt PJ-940 (S3 horizontal) (2)
- Filter cartridge for FRL1 (2)
- Fuse kit (10× each of 1A, 2A, 4A, 10A 5×20 mm)
- Cabinet door key (2)
- ATC pull stud, ISO30 type B (8)
- ER32 collet kit, Ø 3 to Ø 20 mm (1)

12. Sign-off

Installation is considered complete when:

1. Anchors are cured.
2. All utilities pass §4 / §5 / §6 checks.
3. Acceptance program from §9 has been run and signed.
4. Operator training §10 has been delivered and signed.
5. The customer has signed the handover certificate at the HMI.

The warranty start date is the date on the signed handover certificate.